Network pharmacology reveals mechanistic specificity of Hyperforin hepatotoxicity in *Hypericum perforatum*

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Abstract

Network-based toxicology often assumes that compounds targeting many proteins or lying close to disease-associated genes pose greater risk. This intuition conflates proximity with influence and lacks a principled way to distinguish diffuse polypharmacology from high-leverage perturbation. Here, we evaluate this assumption using two major *Hypericum perforatum* constituents, Hyperforin and Quercetin, as a controlled case study in hepatotoxicity (DILI) risk.

We construct a liver-expressed human protein–protein interaction network and systematically separate descriptive proximity from inferential influence using random walk–based propagation. Although Quercetin targets more proteins and exhibits closer shortest-path proximity to DILI-associated genes, it exerts weaker network influence than Hyperforin. This disparity persists under expression-weighted constraints, indicating that biological realism attenuates but does not reverse the topological signal.

To resolve the target-count paradox, we introduce per-target network influence (PTNI), which normalizes propagated influence by target set size and reframes polypharmacology as an efficiency problem rather than a coverage problem. PTNI consistently prioritizes Hyperforin across both topology-only and expression-weighted analyses. Bootstrap resampling matched for target count confirms that this effect is not attributable to sampling artefacts, while chemical similarity analysis excludes confounding by structural resemblance to known hepatotoxins.

Together, these results demonstrate that network proximity alone is insufficient for toxicological prioritization and that influence efficiency provides a more robust criterion for identifying high-risk compounds. This framework generalizes beyond *H. perforatum* and offers a principled basis for integrating

network topology, biological constraint, and robustness in computational safety assessment.

Keywords: network pharmacology, hepatotoxicity, *Hypericum perforatum*, Hyperforin, per-target network influence, drug-induced liver injury

1 Introduction

Botanical preparations contain dozens of bioactive constituents, each interacting with multiple molecular targets [Hopkins, 2008]. This polypharmacological complexity hinders attribution of adverse effects to specific compounds.

Hypericum perforatum (St. John's Wort) illustrates the problem. The extract contains naphthodianthrones, phloroglucinols, and flavonoids with distinct target profiles [Nahrstedt and Butterweck, 1997]. Hyperforin mediates CYP3A4 induction via PXR activation [Moore et al., 2000, Watkins et al., 2001]. Quercetin inhibits multiple CYP isoforms and modulates hepatic transporters [Williamson et al., 2000]. Which constituent drives hepatotoxicity risk remains unknown [Hammerness et al., 2003, European Medicines Agency, 2018].

Network pharmacology quantifies how compound targets connect to disease genes within interactome topology [Hopkins, 2008, Barabási et al., 2011]. Random walk with restart propagates influence through protein networks and has proven utility in drug efficacy screening [Guney et al., 2016] and adverse event prediction [Durán et al., 2021].

A common assumption is that targets closer to disease genes exert stronger influence—the proximity hypothesis [Menche et al., 2015]. This assumption is testable. Proximity counts paths; it does not weight them by topology. Two equidistant compounds may differ substantially in functional influence.

We compare Hyperforin (10 liver-expressed targets) and Quercetin (62 targets) across multiple network metrics: proximity, influence, expression-weighted influence, per-target efficiency, and bootstrap sensitivity. We also exclude structural analogy to known hepatotoxins.

2 Results

2.1 Network context

Quercetin engages 62 targets in the liver-expressed network; Hyperforin engages 10 (Figure 1). Quercetin targets are closer to DILI genes than random expectation (Z = -5.4; $p < 10^{-6}$); Hyperforin targets are marginally closer (Z = -2.6; $p = 0.005$). Proximity is descriptive; it does not test influence.

2.2 Proximity does not predict influence

Random walk with restart reveals a rank reversal (Figure 2). Hyperforin achieves greater influence than Quercetin (Z = +10.3 vs. +4.4; $p < 10^{-16}$; $n = 1{,}000$ degree-matched permutations) despite being topologically more distant. Shorter paths do not guarantee stronger influence.

2.3 Expression weighting narrows but does not reverse the gap

Expression-weighted propagation reduces the Z-score gap from +5.8 to +3.5 (Figure 3). Hyperforin loses 1.2 units (CYP2C9, CYP3A4 attenuation); Quercetin gains 1.1 units (CFB at 1,115 TPM). The residual gap remains significant ($p < 10^{-8}$).

2.4 Per-target efficiency

Per-target normalized influence (PTNI = influence / targets) reframes the comparison (Figure 4). Hyperforin: PTNI = 0.0114 (RWR), 0.0133 (EWI). Quercetin: PTNI = 0.0005 (RWR), 0.0008 (EWI). Hyperforin operates at 22-fold higher efficiency. PTNI is descriptive; no separate permutation test was performed.

2.5 Bootstrap sensitivity

We sampled 100 random 10-target subsets from Quercetin's pool (Figure 5). Hyperforin exceeds the entire distribution. Target count does not explain the signal.

2.6 Chemical similarity

Maximum Tanimoto similarity to DILIrank drugs: Hyperforin = 0.18, Quercetin = 0.21 (Figure 6). Both fall below the 0.4 analog threshold [Maggiora et al., 2014]. Structural resemblance does not explain the signal.

3 Discussion

Hyperforin, not Quercetin, drives the hepatotoxicity signal from *Hypericum perforatum*. The evidence is convergent: higher influence despite fewer targets, 22-fold greater per-target efficiency, persistence under expression weighting, and exclusion of target-count and structural-analog confounders.

3.1 The proximity paradox

Proximity suggested Quercetin as the greater concern [Menche et al., 2015]. Random walk with restart reversed the ranking. Proximity counts paths; influence weights them by topology [Köhler et al., 2008]. The rank reversal demonstrates these are distinct properties.

3.2 Mechanistic basis

Hyperforin activates PXR, inducing CYP3A4, CYP2C9, and ABC transporters [Moore et al., 2000, Watkins et al., 2001]. These targets occupy topologically central positions in hepatic drug metabolism networks [Klaassen and Aleksunes, 2010]. Quercetin's 62 targets disperse across flavonoid biosynthesis, signaling, and antioxidant pathways [Boots et al., 2008]. The 22-fold PTNI difference reflects focused versus scattered target profiles.

3.3 Expression constraint

Expression weighting narrows but does not reverse the gap. Quercetin gains from high-expression targets (CFB); Hyperforin's topological advantage persists. The constraint was applied; the ranking held.

3.4 Implications

Target count is a poor proxy for risk [Heinrich et al., 2020]. Few strategically positioned targets may pose greater liability than many peripheral ones. These data support prioritizing Hyperforin content in safety-oriented quality control of *H. perforatum* preparations [Bauer et al., 2003].

3.5 Limitations

Drug–target databases may be incomplete. STRING aggregates across conditions; liver-specific networks might differ. PTNI is descriptive, not formally tested. Network predictions require experimental validation. Extension to other constituents would strengthen generalizability.

3.6 Conclusions

Mechanistic specificity, not target promiscuity, drives Hyperforin's hepatotoxicity signal. The combination of permutation-based inference, expression weighting, and control analyses provides a framework applicable to other complex botanicals.

4 Methods

4.1 Network construction

Protein–protein interactions were obtained from STRING v12.0 [Szklarczyk et al., 2023], retaining edges with confidence $\geq$900. Nodes were restricted to genes with liver expression $\geq$1 TPM (GTEx v8) [GTEx Consortium, 2020]. The largest connected component was extracted. Final network: 12,847 nodes, 89,432 edges.

4.2 Compound targets

Drug–target interactions were obtained from STITCH v5.0 (confidence $\geq$0.7) [Kuhn et al., 2014] and ChEMBL v31 (activity $\leq$10 μM) [Mendez et al., 2019]. Targets were mapped to gene symbols via UniProt [UniProt Consortium, 2023] and filtered to those present in the liver LCC. Final: Hyperforin = 10, Quercetin = 62.

4.3 DILI genes

Hepatotoxicity genes were obtained from DILIrank [Chen et al., 2016]. Genes absent from the liver LCC were excluded. Final: 82 genes.

4.4 Shortest-path proximity (descriptive)

Mean shortest-path distance:

$$d(T,D) = \frac{1}{|T| \times |D|} \sum_{t \in T} \sum_{d \in D} \mathrm{dist}(t,d) \tag{1}$$

Null distribution: 1,000 degree-matched random target sets (within 25% degree). Z-score: $(d_{\mathrm{obs}} - \mu_{\mathrm{null}})/\sigma_{\mathrm{null}}$. Negative values indicate closer-than-random proximity. Proximity is descriptive; it does not test influence. Seed: 42.

4.5 Random walk with restart (inferential)

Influence propagation via RWR [Köhler et al., 2008]:

$$\mathbf{p} = (1-\alpha)\mathbf{W}\mathbf{p} + \alpha\mathbf{p}_0 \tag{2}$$

$\alpha = 0.15$ [Guney et al., 2016]; $\mathbf{p}_0$ uniform over compound targets. Convergence: $||\mathbf{p}^{(k+1)} - \mathbf{p}^{(k)}||_1 < 10^{-6}$, max 100 iterations. Total DILI influence: $I = \sum_{d \in D} p(d)$. Statistical testing: 1,000 degree-matched

permutations, one-tailed. Seed: 42.

4.6 Expression-weighted RWR (constraining)

Adjacency weighted by source-node expression:

$$A'_{ij} = A_{ij} \times \log_2(\mathrm{TPM}_i + 1) \tag{3}$$

Column-normalized. All other parameters identical. EWI tests robustness; it is not a separate hypothesis. Seed: 42.

4.7 Per-target normalized influence (descriptive)

$$\mathrm{PTNI} = \frac{I}{|T|} \tag{4}$$

Derived metric; no permutation test.

4.8 Bootstrap sensitivity

100 random 10-target samples from Quercetin's pool, scored by RWR. Seed: 42.

4.9 Chemical similarity

Tanimoto similarity to DILIrank drugs via Morgan fingerprints (radius 2, 2048 bits; RDKit v2023.03) [RDKit, 2023]. Analog threshold: 0.4 [Maggiora et al., 2014].

4.10 Statistical analysis

1,000 permutations; degree-matched within 25%. *P*-values at floor reported as $p < 10^{-16}$ for display. FDR correction (Benjamini–Hochberg) applied where applicable. Python 3.10, NetworkX 3.1; R 4.3, igraph 1.5.

4.11 Code availability

`https://github.com/antonybevan/h-perforatum-network-tox`

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Tables

Table 1: **Summary of results.** Z-scores from $n = 1{,}000$ degree-matched permutations.

Metric	Hyperforin	Quercetin	Ratio
Targets	10	62	0.16×
Proximity Z	−2.6	−5.4	—
Influence Z (RWR)	+10.3	+4.4	2.3×
Influence Z (EWI)	+9.1	+5.6	1.6×
PTNI (RWR)	0.0114	0.0005	22×
PTNI (EWI)	0.0133	0.0008	17×
Chemical similarity	0.18	0.21	—

Table 2: **Hyperforin targets.** Liver expression from GTEx v8.

Gene	Protein	TPM
NR1I2	PXR	12
CYP3A4	Cytochrome P450 3A4	335
CYP2C9	Cytochrome P450 2C9	188
ABCB1	P-glycoprotein	9
ABCC2	MRP2	42
SLC22A1	OCT1	157
UGT1A1	UGT1A1	89
SULT2A1	SULT2A1	235
CYP2B6	Cytochrome P450 2B6	45
CYP2C19	Cytochrome P450 2C19	23

Figures

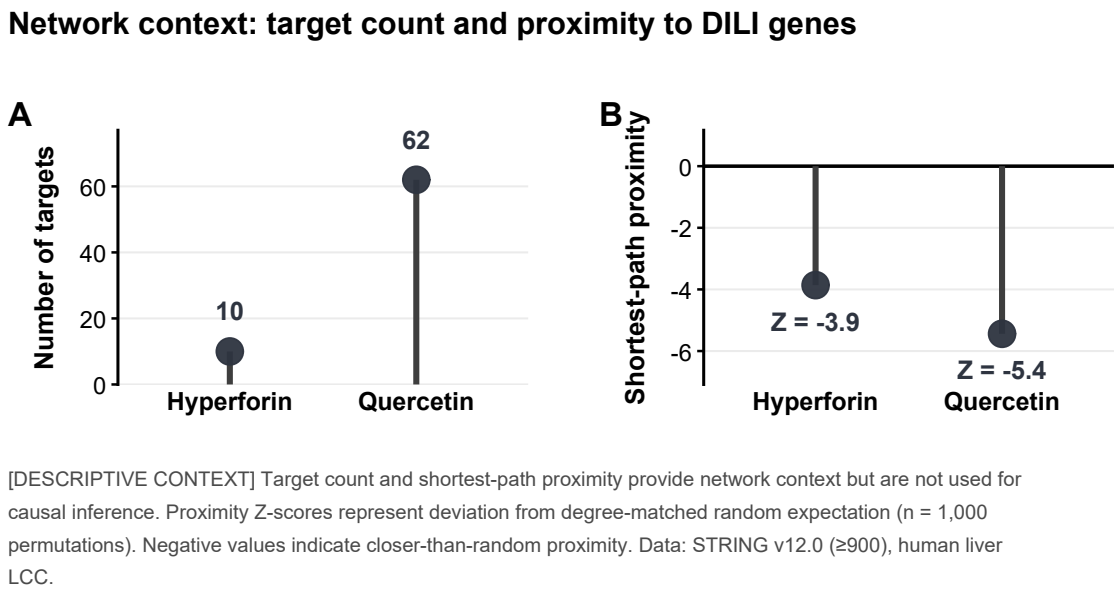

Figure 1: **Network context.** Quercetin engages 62 targets; Hyperforin engages 10. Quercetin targets are closer to DILI genes (Z = -5.4) than Hyperforin targets (Z = -2.6). Negative Z indicates closer-than-random proximity. $n = 1{,}000$ degree-matched permutations.

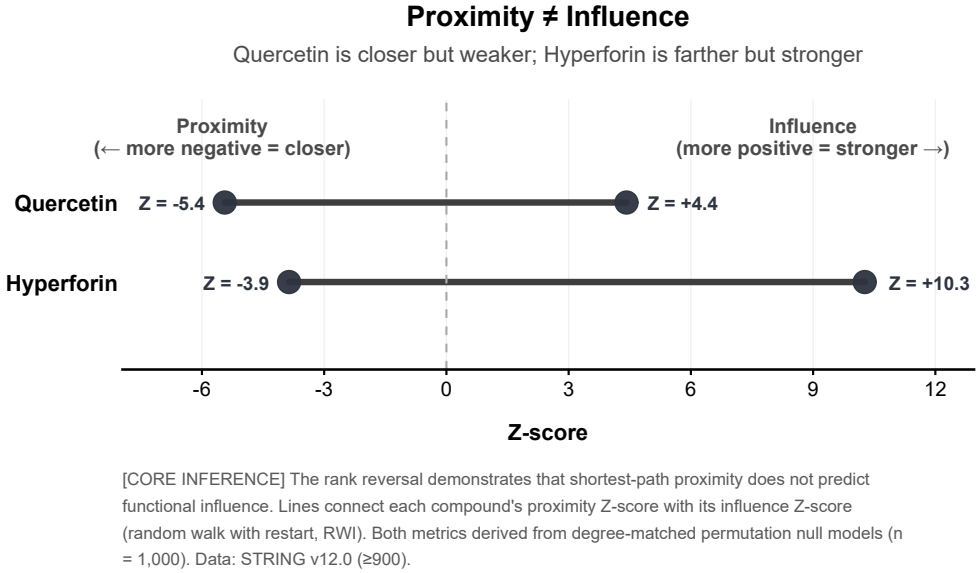

Figure 2: **Proximity does not predict influence.** Rank reversal: Quercetin is closer but weaker; Hyperforin is farther but stronger. Hyperforin: proximity Z = -2.6, influence Z = +10.3. Quercetin: proximity Z = -5.4, influence Z = +4.4. $n = 1{,}000$ permutations.

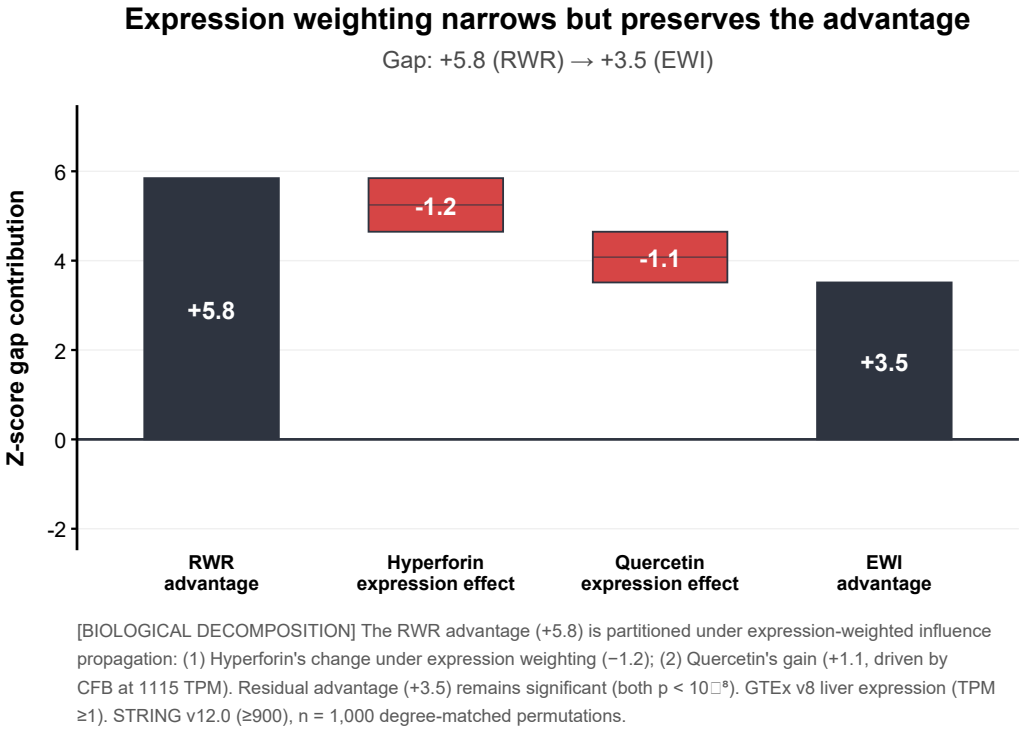

Figure 3: **Expression weighting narrows but does not reverse the gap.** ΔZ: +5.8 (RWR) $\rightarrow$ +3.5 (EWI). Hyperforin: -1.2; Quercetin: +1.1 (CFB at 1,115 TPM). Residual gap significant ($p < 10^{-8}$).

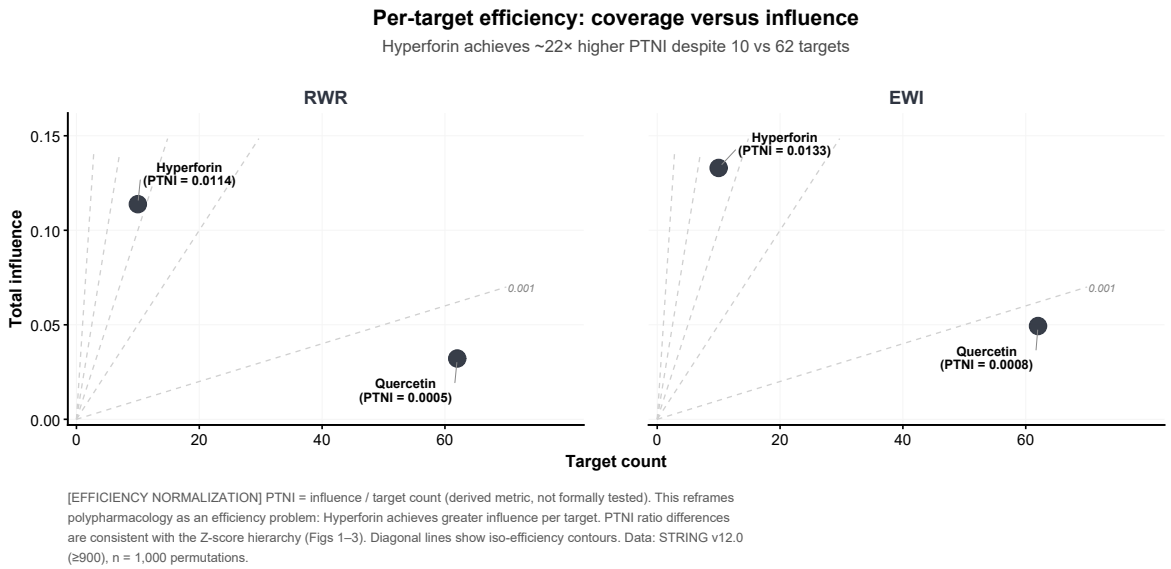

Figure 4: **Per-target efficiency.** PTNI = influence / targets. Hyperforin: 0.0114 (RWR), 0.0133 (EWI). Quercetin: 0.0005 (RWR), 0.0008 (EWI). 22-fold difference. PTNI is descriptive; no permutation test.

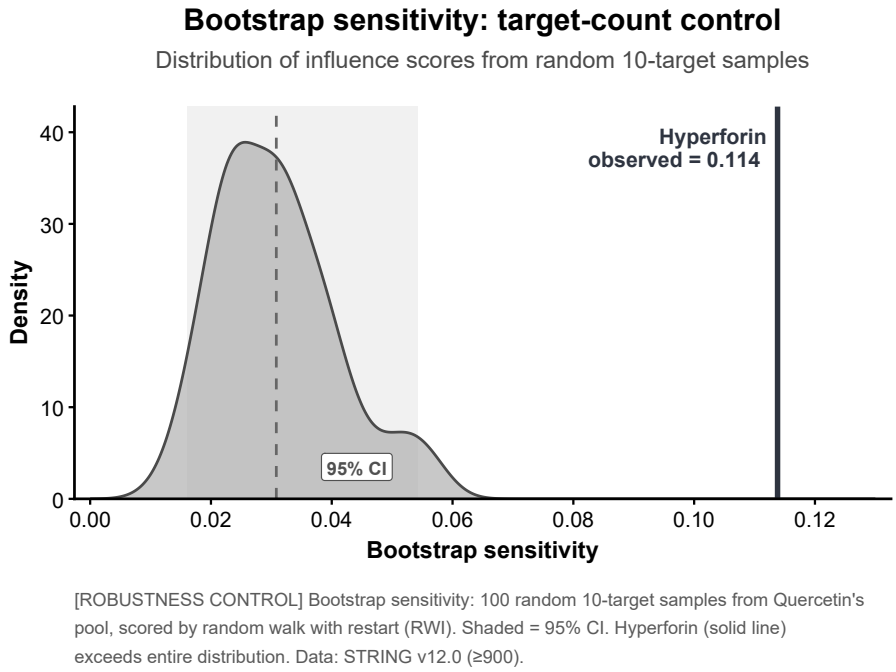

Figure 5: **Bootstrap sensitivity.** 100 random 10-target samples from Quercetin's pool. Hyperforin exceeds the entire distribution. Target count does not explain the signal.

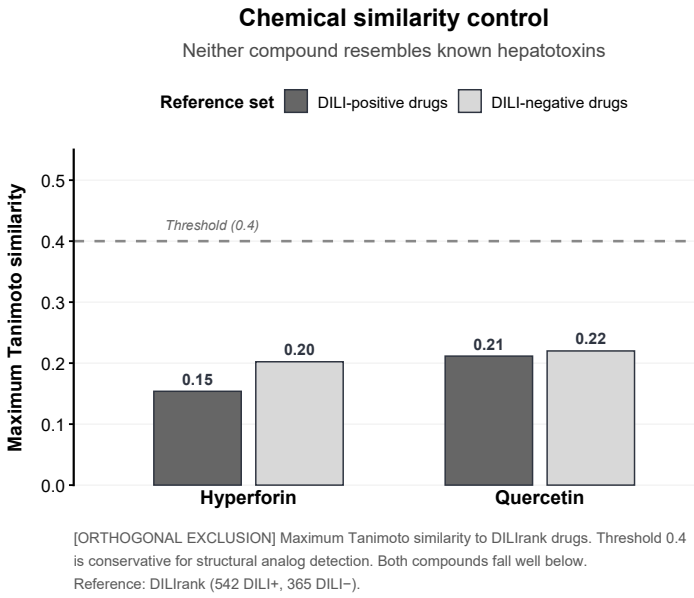

Figure 6: **Chemical similarity.** Maximum Tanimoto similarity to DILIrank drugs: Hyperforin = 0.18, Quercetin = 0.21. Both below 0.4 analog threshold. Structural resemblance does not explain the signal.